Mohith Munagala

DevOps / Cloud Engineer

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PROFILE SUMMARY:

- Designed, implemented, and maintained highly available cloud infrastructure across **AWS and Microsoft Azure** using **Amazon ECS Fargate, EKS, AKS, Docker, Kubernetes, Application Load Balancers (ALB), Auto Scaling, and multi-environment deployments**, ensuring scalable, secure, and production-ready enterprise platforms.
- Engineered secure cloud networking by designing **AWS VPCs, Public/Private Subnets, NAT Gateways, Route Tables, Internet Gateways, Security Groups, IAM, RBAC, and Azure Key Vault**, strengthening infrastructure security and optimizing network performance.
- Built and automated enterprise **CI/CD pipelines** using **GitHub Actions, Jenkins, GitLab CI/CD, Azure DevOps, and Argo CD**, enabling reliable application delivery, infrastructure provisioning, automated testing, and zero-downtime deployments.
- Developed reusable **Infrastructure as Code (IaC)** solutions using **Terraform, AWS CloudFormation, AWS CDK, Azure Bicep, and Helm**, delivering consistent, scalable, and repeatable cloud infrastructure deployments across multiple environments.
- Deployed and managed **containerized microservices** using **Docker, Kubernetes (EKS, AKS, On-Prem), and Amazon ECS Fargate**, implementing GitOps, autoscaling, self-healing, blue-green, and canary deployment strategies for high application availability.
- Implemented enterprise **DevSecOps** practices using **AWS IAM, Azure RBAC, AWS KMS, Azure Key Vault, SonarQube, SSL/TLS, and security policies**, ensuring secure access control, secrets management, compliance, and application security.
- Designed and supported **AI/ML cloud platforms** by deploying **GPU-enabled Kubernetes clusters**, **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)** solutions, **PostgreSQL**, **Neo4j**, and **NVIDIA GPU** infrastructure for enterprise AI workloads.
- Established comprehensive monitoring, logging, and observability solutions using **Prometheus, Grafana, Amazon CloudWatch, Azure Monitor, Splunk, and Datadog**, enabling proactive alerting, root cause analysis, incident management, and performance optimization.
- Automated infrastructure operations using **Python, Bash, Ansible, PowerShell, YAML, and JSON**, improving deployment efficiency, configuration management, system reliability, and reducing manual operational effort.

TECHNICAL SKILLS

- **Cloud Platforms:** AWS (EC2, S3, RDS, Lambda, ECS Fargate, EKS, ECR, VPC, IAM, KMS, CloudWatch, Route 53, ALB, API Gateway, SQS, AWS CDK, CloudFormation), Microsoft Azure (AKS, Azure AD, Azure Key Vault, Azure Monitor, Azure SQL, Azure Bicep)
- **Containerization & Orchestration:** Docker, Kubernetes (EKS, AKS, On-Prem), Amazon ECS Fargate, Helm, OpenShift, Traefik Ingress Controller, Docker Compose
- **Infrastructure as Code (IaC):** Terraform, AWS CloudFormation, AWS CDK, Azure Bicep, ARM Templates, Helm, Ansible
- **CI/CD & DevOps:** GitHub Actions, Jenkins, GitLab CI/CD, Azure DevOps, Argo CD (GitOps), AWS CodeDeploy, Blue/Green & Canary Deployments
- **Scripting & Automation:** Python, Bash, PowerShell, YAML, JSON
- **Monitoring & Observability:** Prometheus, Grafana, Amazon CloudWatch, Azure Monitor, Splunk, Datadog, SonarQube
- **Networking & Security:** AWS VPC, Subnets, NAT Gateway, Route Tables, Security Groups, IAM, Azure RBAC, AWS KMS, Azure Key Vault, HTTPS, SSL/TLS, OAuth 2.0
- **Databases:** PostgreSQL, MySQL, Amazon RDS, DynamoDB, Azure SQL, Elasticsearch, Neo4j
- **AI / ML Platforms:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Knowledge Graphs, Ollama, GPU Infrastructure, NVIDIA H200
- **Serverless & Messaging:** AWS Lambda, API Gateway, Amazon SQS
- **DevSecOps & Compliance:** IAM, RBAC, KMS, SonarQube, HIPAA, GDPR
- **Agile & Platforms:** Git, GitHub, GitLab, Linux, Nginx, Agile/Scrum

WORK EXPERIENCE

AWS DevOps Engineer | Amazon

Remote, USA | Apr 2026 – Present

Key Responsibilities:

- Architected and implemented secure, highly available AWS cloud infrastructure using **Amazon ECS Fargate, VPC, IAM, Amazon RDS, Amazon S3, Application Load Balancer (ALB), CloudWatch, Route Tables, NAT Gateways, and Security Groups**, supporting scalable multi-environment enterprise applications.
- Designed and automated end-to-end **CI/CD pipelines** using **GitHub Actions, Docker, Amazon ECR, Amazon ECS Fargate, and AWS CDK**, enabling automated build, image publishing, infrastructure provisioning, application deployment, rollback, and release management.
- Developed reusable **Infrastructure as Code (IaC)** solutions using **AWS CDK (TypeScript)** for networking, ECS services, databases, monitoring, secrets management, and security configurations, ensuring standardized and consistent infrastructure deployments.
- Deployed and managed **containerized microservices** using **Docker, Amazon ECS Fargate, Application Load Balancer (ALB), ECS Task Definitions, Auto Scaling Policies, and SSL integration**, delivering highly available, secure, and production-ready application environments.
- Implemented secure cloud architecture by integrating **AWS IAM, AWS KMS, AWS Secrets Manager, Systems Manager Parameter Store, Security Groups, and VPC networking**, strengthening access control, secrets management, and enterprise security compliance.
- Built centralized **monitoring and observability** solutions using **Amazon CloudWatch Dashboards, CloudWatch Alarms, ECS Metrics, ALB Metrics, RDS Monitoring, and centralized logging**, enabling proactive alerting, performance optimization, and incident management.
- Developed **Python and Bash automation** scripts to streamline infrastructure provisioning, deployment workflows, environment configuration, operational tasks, and production support, improving deployment consistency and reducing manual effort.

Environment: *AWS (Amazon ECS Fargate, EC2, VPC, IAM, Amazon S3, Amazon RDS, Amazon ECR, Application Load Balancer (ALB), CloudWatch, Route 53, API Gateway, SQS, AWS CDK, AWS KMS, AWS Secrets Manager, Systems Manager Parameter Store), Docker, GitHub Actions, TypeScript, PostgreSQL, Python, Bash, CI/CD, Infrastructure as Code (AWS CDK), Agile/Scrum.*

Azure DevOps Engineer | Caterpillar

Peoria, IL, USA | Sep 2025 – Mar 2026

Key Responsibilities:

- Architected and deployed **GPU-enabled Kubernetes (On-Prem)** infrastructure supporting enterprise **AI platforms**, enabling highly available and scalable **Large Language Model (LLM)** inference workloads for production environments.
- Designed and automated infrastructure provisioning and deployment workflows using **Terraform, Azure Bicep, ARM Templates, Ansible, and Azure Key Vault**, ensuring consistent, secure, and repeatable infrastructure deployments.
- Built and implemented automated **CI/CD pipelines** using **GitHub Actions** and **Argo CD (GitOps)**, enabling continuous integration, automated testing, image versioning, and application deployment across multiple environments.
- Deployed and managed **containerized microservices** using **Docker, Kubernetes, Helm, OpenShift, and Traefik Ingress Controller**, implementing autoscaling, self-healing, secure ingress, and highly available application deployments.
- Designed and implemented enterprise **AI/ML solutions** by integrating **Retrieval-Augmented Generation (RAG)** pipelines with **Large Language Models (LLMs), PostgreSQL, Neo4j, and GPU infrastructure**, enabling scalable and context-aware AI applications.
- Configured secure authentication and access management using **Azure Active Directory, Azure RBAC, and Azure Key Vault**, enforcing enterprise security standards and protecting cloud-native workloads.
- Developed and maintained **Ansible playbooks** for node provisioning, configuration management, patch automation, and infrastructure lifecycle management, improving deployment consistency and reducing manual operational effort.
- Implemented distributed storage and Kubernetes networking using **Longhorn, Traefik, SSL/TLS, and load balancing**, ensuring persistent storage, secure service exposure, and resilient application connectivity.

- Established comprehensive monitoring and observability using **Prometheus, Grafana, Azure Monitor, and GitHub self-hosted runners**, enabling proactive alerting, infrastructure monitoring, performance optimization, and production stability.
- Collaborated with cross-functional Agile teams to perform **performance tuning, capacity planning, root cause analysis (RCA), and infrastructure optimization**, improving reliability, scalability, and operational efficiency across enterprise AI platforms.

Environment: *Microsoft Azure, Azure DevOps (YAML), ARM Templates, Terraform, PowerShell, Azure CLI, AKS, Docker, Helm Charts, Nginx, Ingress Controllers, Azure Active Directory, ADFS, AD Connect, Azure Security Center, Azure Policy, Azure Monitor, Log Analytics, Application Insights, Azure Recovery Services Vault, Azure Backup, Azure Virtual Machines, VNet, VM Scale Sets, Git, Azure Repos, RBAC, SSO, KMS / Key Encryption Keys, CI/CD, IaC, Agile/Scrum, Microservices, Blue/Green Deployments*

Cloud Engineer | British Airways

Newyork, USA | June 2024 – Sep 2025

Key Responsibilities:

- Designed and deployed highly available **AWS EKS** infrastructure using **Terraform, AWS CloudFormation, VPC, Public/Private Subnets, NAT Gateway, IAM, KMS, Route 53, and Application Load Balancer (ALB)**, supporting scalable enterprise applications.
- Built and optimized end-to-end **CI/CD pipelines** using **GitHub Actions, Jenkins, Docker, Helm, and Amazon ECR**, automating application build, testing, container image management, and Kubernetes deployments across multiple environments.
- Deployed and managed **containerized microservices** using **Docker and Kubernetes (Amazon EKS)**, implementing autoscaling, self-healing, rolling updates, and highly available cloud-native application architectures.
- Designed and implemented **RESTful APIs** using **AWS API Gateway** and **AWS Lambda**, enabling secure, scalable, and event-driven application integration for enterprise services.
- Automated cloud infrastructure and operational tasks using **Python (Boto3)** for EKS scaling, Amazon S3 audits, CloudWatch log management, and infrastructure optimization, reducing manual operational effort.
- Implemented comprehensive **monitoring and observability** using **Prometheus, Grafana, Amazon CloudWatch, and Splunk**, enabling proactive alerting, performance monitoring, root cause analysis, and production support.
- Strengthened cloud security by implementing **AWS IAM, AWS KMS, SSL/TLS, Security Groups, and secure VPC networking**, ensuring enterprise security, compliance, and secure application communication.

Environment: *AWS (EKS, EC2, VPC, IAM, KMS, S3, RDS, Lambda, API Gateway, Route 53, Application Load Balancer, CloudWatch, CloudFormation, Amazon ECR), Terraform, Docker, Kubernetes, Helm, GitHub Actions, Jenkins, Python (Boto3), Prometheus, Grafana, Splunk, SonarQube, Git, PostgreSQL, SSL/TLS, CI/CD, Infrastructure as Code (IaC), Agile/Scrum.*

Cloud Engineer | TCS

Hyderabad, India | May 2021 - July 2023

Key Responsibilities:

- Designed and automated AWS cloud infrastructure using **Terraform, AWS CloudFormation, EC2, VPC, S3, RDS, Lambda, IAM, and API Gateway**, provisioning scalable and secure environments through Infrastructure as Code (IaC).
- Built and optimized **CI/CD pipelines** using **GitLab CI/CD, AWS CodeDeploy, Docker, and Kubernetes (Amazon EKS)**, automating application build, testing, deployment, and release workflows across multiple environments.
- Deployed and managed **containerized microservices** using **Docker, Amazon EKS, and Helm**, implementing scalable, highly available, and cloud-native application architectures.
- Architected secure AWS networking using **VPC, Public/Private Subnets, Route Tables, Internet Gateway, Security Groups, IAM, and AWS KMS**, ensuring secure communication and compliance with enterprise cloud security standards.
- Implemented comprehensive monitoring and observability using **Prometheus, Grafana, Amazon CloudWatch, CloudTrail, and Splunk**, enabling proactive alerting, root cause analysis, incident management, and continuous infrastructure optimization.

EDUCATION

Master of Science in Computer Technology

Eastern Illinois University, Charleston, Illinois, USA